Imperial College London
Department of Earth Science and Engineering
MSc Environmental Data Science and Machine Learning

Comparison of Direct and Cascaded High-Resolution Spatial Downscaling of Wildfire Data

Author:
Tsuyu Huang
tsuyu.huang25@imperial.ac.uk

Supervisors:
Jorge Veiras (jorge@artiocarbon.com)
Prof. Anna Korre (a.korre@imperial.ac.uk)

Code Repository:
https://github.com/ese-ada-lovelace-2025/irp-th25

Company:
Artio
www.artiocarbon.com

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Abstract

With climate change, there's an increasing need for a higher-resolution wildfire data for better monitoring and risk assessment. GEO satellites have high-temporal resolution, but lacks spatial resolution for a detailed risk assessment. While LEO satellites provide better spatial resolution, but low temporal resolution to monitor rapid wildfire spread. Existing downscaling models have demonstrated their effectiveness in producing high-resolution remote sensing data. But many still fail to capture the fine details and physically plausible outputs. This project aims to achieve a high-resolution spatial downscaling of wildfire data by integrating additional physical covariates to the model to mitigate implausible outputs. The project explores direct (2km to 30m) and cascaded (2km to 375m to 30m) downscaling pipelines and their effectiveness in producing high-resolution wildfire data. The project compares effectiveness of hybrid attention U-Net (HAUNet) as wildfire downscaling model.

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1 Introduction

In recent years, due to rising popularity in Net-Zero initiatives, carbon investors are increasingly funding carbon reduction and removal projects, such as reforestation (Lambert and Turner, 2024). However, these carbon projects are at risk because wildfires can destroy and release the carbon back into the atmosphere. With climate change, wildfires are becoming increasingly severe and frequent (Jones et al., 2020). This causes economic losses for the insurance industry. For instance, The January 2025 Los Angeles County wildfires set an all-time record, causing $40 billion in insured losses alone from a single event (Dwyer and Nam, 2026). This project, conducted with Artio, aims to address this gap. By providing highly accurate risk assessment models, this research seeks to give better confidence and enable better risk assessment.

High-spatial and high-temporal resolution wildfire data is favourable in risk assessment. However, high-temporal resolution data obtained from geostationary satellites (GEO) have limited spatial resolution, which makes them difficult to use for accurate risk assessment. While low orbit satellites (LEO) offers high-spatial resolution, but low-temporal resolution.

To bridge this gap, existing methods uses Deep Learning (DL) models for downscaling due to their ability in recognising non-linear spatiotemporal patterns. However, the resolution gap between GEO and LEO is significant (2km - 30m). This increases the risk of producing physically implausible or unrealistic outputs, as a single low-resolution (LR) pixel can correspond to different high-resolution (HR) details (Z. Wang et al., 2020). In this project, to mitigate this limitation, the model integrates additional physical information, such as HR topography and landcover to help guide the downscaling process. To further improve downscaling, the project aims to investigate whether cascaded downscaling (2km to 375m to 30m) produces more accurate and better results compared to direct downscaling (2km to 30m). By breaking up the downscaling process, the model trains on a smaller resolution gap, aiming to mitigate error caused by the large resolution difference. This project will compare the cascaded and direct pipeline performance of different downscaling architectures, including hybrid attention U-Net.

2 Current State of Remote Sensing Downscaling

Traditional downscaling methods often fail to capture the complex non-linear patterns in remote sensing data. Recently, Deep learning methods have shown to outperform these approaches (Sdraka et al., 2022). CNN and U-Net architectures have been widely used for their ability to capture spatial features. While GAN-based methods have been used where sharpness is prioritised over pixel-wise accuracy. More recently, transformer-based models use long-range attention to capture dependencies beyond a convolution's receptive field (J. Wang et al., 2025), and diffusion-based approaches have showed strong generative ability (Y. Wang et al., 2023).

However, many downscalingarchitectures are optimised for texture-rich images. They are not well-suited to the distinct structure of active wildfire data where the target is very sparse. This creates two challenges: severe class imbalance, which biases pixel-wise reconstruction losses and convolutional decoders (Qi et al., 2026); and an ill-posed one-to-many mapping, where a single low-resolution pixel have many possible high-resolution configurations. Architectures are needed that can both preserve fire signal against a dominant background and recover coherent fire-front structure.

Existing literature is dominated by direct downscaling across the full resolution gap. Cascaded approaches exist e.g. AMCN-Geoi-DBN (Jing et al., 2022) and CLIT (Chen et al., 2023) but no prior work evaluates cascaded downscaling for wildfire detection, nor explores the trade-off between the reduced per-stage resolution gap and the accumulation of error across stages. The objectives are:

1. Develop a domain-specific, multi-stage framework for wildfire downsclaing.

2. Compare the performance of HAUNet compared to baseline autoencoder in terms of accuracy and quality of the downscaled predicted fires.

3. Evaluate the trade-offs of cascaded and direct downscaling and its effectiveness in improving the accuracy of wildfire downscaling.

3 Data and Materials

Table 1 shows the data sources used for this project. The specifications for the VIIRS, GOES, and Landsat data are outlined in Table 2.

The GOES-16/17 data were obtained via public Google Cloud Platform (GCP) buckets (NOAA, 2017, 2018). DEM, and ESA WorldCover are obtained via Google Earth Engine (GEE) Python API (Gorelick et al., 2017) and exported directly to Google Cloud Storage (GCS). The Landsat-8 fire masks are derived by de Almeida Pereira et al., 2021 and obtained from the author's public dataset (hosted on Google Drive) and re-uploaded to GCS for pipeline consistency. VIIRS fire detections are obtained from NASA FIRMS archive. Manifests, split assignments and model checkpoints are also stored in GCS.

The data was processed so that all regions of interest (ROIs) are dynamically reprojected to their localised UTM zone based on centroid coordinates. Detections from the NASA VIIRS S-NPP archive were filtered for "nominal" and "high" confidence points. Detections are clustered into 7680 × 7680 m grid cells within a 15-minute temporal window of peak activity. VIIRS grid cell pixels record the peak Fire Radiative Power (FRP). Due to extreme class imbalance of the target fire datasets (mean fraction of positive target pixels per patch are 0.85% for VIIRS, 0.06% for Landsat), stratified sampling was applied across temporal extent and fire pixel count bins during data splitting. The closest temporal GOES-16/17 scan within a ±60-minute guard window was queried.

GOES-16/17 channels 7 (3.9 μm, shortwave IR) and 14 (11.2 μm, longwave IR) are used as they directly correspond to the spectral bands for fire-detection for their sensitivity to fire radiance. This is similar spectral logic in VIIRS's own active-fire algorithm, which similarly relies on a shortwave (I4, 3.74 μm) and longwave (I5, 11.45 μm) channel pair. (Schmidt et al., 2013)

In addition to the native 3.9 μm (Ch. 7) and 11.2 μm (Ch. 14) GOES brightness temperatures, a Thermal Fire Index (TFI) is computed as additional input channel:

$$TFI = Ch.7 - Ch.14 \tag{1}$$

Explicitly providing the difference reduces the burden on early network layers to discover a known, physically meaningful relationship, aiming to save model capacity and improving convergence.

The dataset is split into training/validation/testing (80/10/10). This split utilises stratified random sampling across four quantile bins based on the active fire pixel count per patch for balanced representation of fire sizes. Weighted random sampling is applied during training to address class imbalance, where sample weights w_i for each patch i were assigned based on the count of positive fire pixels n_i:

$$w_i = \max(n_i, 1) \tag{2}$$

where n_i is extracted from the patch metadata manifests.

Inputs are scaled prior to model ingestion. The GOES and DEM channels undergo Z-score standardisation using mean and standard deviation parameters precomputed exclusively from the training split. DEM covariates were normalised by dividing by 1000 (converting to km). The landcover data is also one-hot encoded into 11 channels, each representing a landcover class.

For the targets, the continuous VIIRS FRP values are binarised into active fire masks. Landsat targets are already binarised. Depending on the experimental configuration, thes are subsequently transformed using either spatial dilation via a 3×3 max pooling operation or conversion into continuous Gaussian heatmaps.

Measured on the held-out test split, the fire-pixel prevalence was 0.008% (VIIRS) / X% (Landsat), confirming the extreme class imbalance

Table 1: Data Sources and Specifications

Data Source	**Res.**	**Role / Type**	**Bands Used**	**Description / Pipeline Context**
Landsat-8 Fire Masks	30 m	Target / Ground Truth	Active Fire Mask generated by de Almeida Pereira et al., 2021	High-resolution active binary fire mask (256×256 px).
VIIRS S-NPP	375 m	Intermediate Target / Input in Cascaded Pipeline	I4 (3.74 μm, Active Fire)	Active fire detection product (NASA FIRMS) in Fire Radiative Power (FRP) used for cascaded downscaling.
GOES-16/17 (ABI)	2000 m	Coarse Fire Input	Ch. 7 (3.9 μm), Ch. 14 (11.2 μm)	High-temporal-frequency geostationary data in Brightness Temperature (BT).
Copernicus DEM	30 m	Covariate (Static)	DEM	Digital Elevation Model to capture terrain effects on fire spread.
ESA WorldCover	10 m	Covariate (Static)	All Landcover types (11)	Landcover maps.

4 Methods

4.1 HAUNet (Hybrid Attention U-Net)

Among CNN-based approaches, HAUNet (J. Wang et al., 2023) uses hybrid attention to capture both local fine detail and broader contextual structure across multiple scales. HAUNet was selected for its ability to capture local extreme hotspots in near-zero background, which standard U-Nets might smooth out.

Table 2: Data Pipeline Configuration Specifications

Parameter / Constant	Value	Description
VIIRS Geographic Bounds	$[-170.0, -30.0]$	Geographic longitude limits for the Americas
VIIRS Date Range	2020-01-01 to 2020-12-31	Temporal query window for VIIRS detections
Landsat Geographic Bounds	North and South America	Locations of interest for GOES detections
Landsat Date Range	Aug and Sep 2020	Temporal window for Landsat detections (de Almeida Pereira et al., 2021)
Minimum Landsat Active Pixels	10	Minimum active Landsat fire pixels required to retain a patch
GOES Temporal Guard	60 minutes	Max allowed temporal gap between target time and GOES scan
GOES Native Scale	2000 m	Native spatial grid spacing of GOES ABI channel data
Landsat Grid Scale	30 m	Spatial resolution for Landsat processing branch
VIIRS Grid Scale	375 m	Spatial resolution for VIIRS active fire detections
Landsat Dimensions	256×256 px	Target dimensions for Landsat patches
VIIRS Dimensions	20×20 px	Target spatial dimension for VIIRS rasters (7680/375)
GOES Dimensions	4×4 px	Target spatial dimension for GOES rasters (7680/2000)

4.1.1 Hybrid Attention Mechanisms

- **Channel Attention Module (CEM):** Computes channel dependencies with $O(C^2)$ complexity. Given feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ are generated via 1×1 convolutions followed by 3×3 depthwise convolutions. The channel attention map $\mathbf{A}_c$ is derived as:

$$\mathbf{A}_c = \text{Softmax}\left(\text{ReLU}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\alpha}\right)\right)\mathbf{V} \tag{3}$$

  where α is a learnable scaling parameter.

- **Spatial-Channel Attention (S_CEM):** Parallel spatial attention branch operating along transposed dimensions $\mathbf{Q}_s, \mathbf{K}_s, \mathbf{V}_s \in \mathbb{R}^{B \times N_{head} \times HW \times C}$. The full spatial attention step scales quadratically with spatial resolution as $O((HW)^2)$.

4.1.2 Full-Resolution Covariate Pyramid Branch

Because full spatial attention in S_CEM imposes an $O((HW)^2)$ memory footprint that causes out-of-memory errors on full-resolution inputs, the inputs (256×256) are downsampled by factor $D = 4$ (i.e. $256/4 = 64$) before passing to the model.

The parallel Covariate Pyramid Encoder was introduced because downsampling covariates alongside the fire input discards the HR detail at the target 30m scale. Processing covariates at full resolution in a separate branch, then progressively fusing into the main decoder path. This branch processes full-resolution covariate features using Windowed Spatial-Channel Attention Blocks (Windowed-S_CEM) (Liu et al., 2021) for computational efficiency.

For a feature map partitioned into non-overlapping local windows of size $P \times P$ ($P = 8$), Windowed-S_CEM computes spatial attention independently within each window. The window is shifted sequentially with shift size $P/2$. This keeps the spatial attention complexity to $O(P^4)$ per window, independent of overall spatial input dimensions.

4.1.3 Multi-Scale Refinement and Feature Fusion

The main encoder-decoder path produces low-resolution feature representations (64×64). The final refinement stage produces the final high-resolution output. At each stage:

1. Low-resolution features are upsampled via bilinear interpolation followed by 3×3 convolutions.
2. Upsampled features are concatenated with matching scale representations extracted from the Covariate Pyramid Encoder.
3. A 3×3 fusion convolution maps concatenated features back to channel dimension.

4.1.4 Architecture Selection Rationale

HAUNet was selected as one of the architectures for the following reasons:

- It uses dual attention: channel attention to capture global information across feature channels and spatial attention to capture spatial relationships.
- Its CIM used as skip connections. Skip connections used in conventional U-Net concatenate features at matching spatial resolution, but this assume they are semantically comparable. CIM addresses this by explicitly bridging semantic and resolution gaps between encoder outputs at different scales prior to fusion.

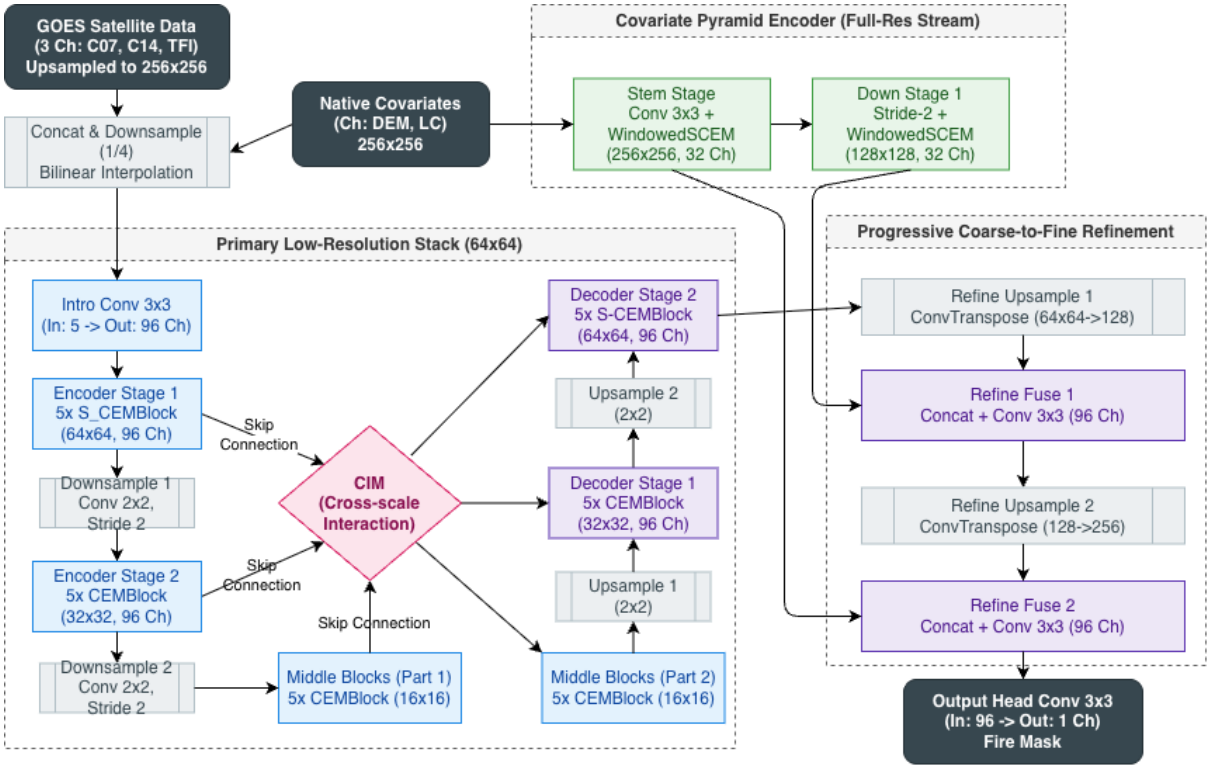


Figure 1: Architecture overview of the wildfire-adapted HAUNet. The full-resolution stream processes native covariates, which are progressively fused via refinement blocks with the low-resolution stack after CIM.

- Both the autoencoder baseline and HAUNet share the same underlying encoder-decoder-with-skip-connections structure. This allows the comparison between the two models to directly isolate the contribution of HAUNet's attention mechanisms and skip-connection design.

4.2 Autoencoder Baseline

The baseline architecture follows the autoencoder proposed by Badhan et al., 2024, adapted to this project to concatenate the covariates with the fire data inputs, and outputs are fire mask segmentation rather than continuous BT values. The autoencoder follows a standard U-Net-style structure, shown in Figure 2.

This baseline is trained under identical conditions to HAUNet (same loss, optimiser, data splits, and covariate set) so that any performance gap are attributable to architectural differences. It establishes a lower bound against which HAUNet's improvement can be quantified, and isolating whether performance gains stem specifically from attention and full-resolution covariate processing, or would be achievable with a standard convolutional model.

4.3 Loss Functions

4.3.1 Focal Dice Loss

Active fire pixels represent a small fraction of each patch, resulting in extreme class imbalance. To address this, Focal Dice Loss was used. It combines Focal Loss (which down-weights background pixels via $(1 - p_t)^\gamma$) with Dice Loss, which optimises for spatial region overlap:

$$\mathcal{L}_{\text{FD}} = w_f \cdot \mathcal{L}_{\text{focal}} + w_d \cdot \mathcal{L}_{\text{dice}} \tag{4}$$

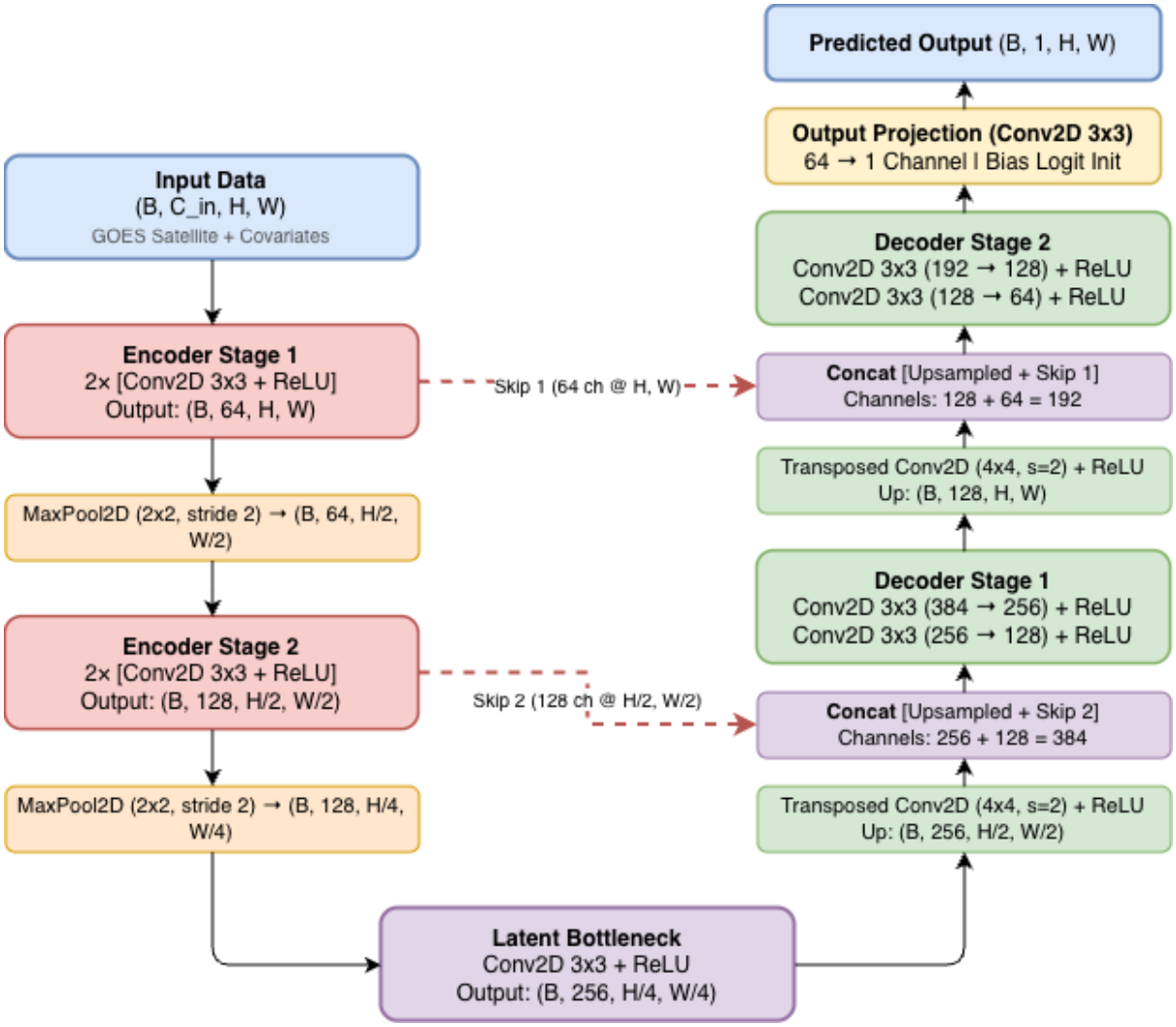


Figure 2: Architecture overview of the wildfire-adapted autoencoder baseline based on Badhan et al., 2024.

where w_f and w_d are weights of the losses.

Focal Loss (Lin et al., 2018) applies $(1 - p_t)^\gamma$ to focus training on sparse positives:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \tag{5}$$

where α_t is the balancing factor, p_t is the predicted probability for the ground-truth class, and $p_{i,j} = P(y_{i,j} = 1)$ is the estimated probability for the positive class. For a ground-truth label $y_{i,j} \in \{0, 1\}$:

$$p_t = \begin{cases} p_{i,j} & \text{if } y_{i,j} = 1 \\ 1 - p_{i,j} & \text{if } y_{i,j} = 0 \end{cases}, \quad \alpha_t = \begin{cases} \alpha & \text{if } y_{i,j} = 1 \\ 1 - \alpha & \text{if } y_{i,j} = 0 \end{cases} \tag{6}$$

$\alpha \in [0, 1]$ balances positive and negative pixel contributions, and $\gamma \geq 0$ controls the rate at which pixels are down-weighted.

Dice Loss (Milletari et al., 2016) tries to maximise the spatial overlap:

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \sum_{i,j} p_{i,j} y_{i,j} + \epsilon}{\sum_{i,j} p_{i,j} + \sum_{i,j} y_{i,j} + \epsilon} \tag{7}$$

where ϵ is a smoothing term preventing division by zero.

4.4 Computational Infrastructure

Model development and training were conducted on a GCP Vertex AI Workbench instance equipped with an NVIDIA L4 GPU, 23.7 GB VRAM, 32 GB RAM, 8 vCPUs. Data generation and preprocessing were run as separate CPU-only jobs querying GCS directly.

4.5 Experimental Design

To evaluate the contribution of the Covariate Pyramid Encoder, HAUNet was assessed under the following ablations, in addition to the primary direct-vs-cascaded pipeline comparison:

1. with and without the Covariate Pyramid Encoder
2. with and without DEM as an input covariate
3. with and without land cover as an input covariate.

In the cascaded pipeline, Stage 1 (GOES to VIIRS) is trained on GOES/VIIRS pairs to predict a VIIRS-resolution (375m) fire mask from the GOES input. Stage 2 (VIIRS to Landsat) is intended to further downscale this to Landsat resolution (30m). However, VIIRS and Landsat overpasses times do not overlap, so matching VIIRS/Landsat pairs are not directly available in the same way GOES/VIIRS pairs are. To address this temporal mismatch, Stage 2 is instead trained using the trained Stage 1 model's predictions on GOES/Landsat dataset as inputs, rather than ground-truth VIIRS observations. For each GOES/Landsat pair, the Stage 1 model produces a predicted VIIRS-resolution fire mask from the temporally matched GOES input, and this prediction is paired with the corresponding Landsat ground truth to train Stage 2. This ensures both stages of the cascaded pipeline operate on inputs available at inference time (i.e. Stage 2 never has access to real VIIRS observations at test time either), avoiding a train/inference mismatch that would occur if Stage 2 were trained on ground-truth VIIRS data unavailable in a genuinely cascaded operational setting.

Each experiment (architecture, covariate set, pipeline stage) was defined by YAML configuration files specifying model hyperparameters, data paths, loss weights, and training settings for reproducibility.

4.6 Evaluation Metrics

- **IOU (Intersection over Union):** Measures the overlap between predicted and ground truth fire masks:

$$\text{IoU} = \frac{\sum_{i,j} \tilde{Y}_{i,j} Y_{i,j}}{\sum_{i,j} \left(\tilde{Y}_{i,j} + Y_{i,j} - \tilde{Y}_{i,j} Y_{i,j} \right) + \epsilon} \tag{8}$$

- **Precision and Recall:** Reported alongside IoU to capture asymmetric errors in fire detection.

$$\text{Precision} = \frac{TP}{TP + FP + \epsilon}, \quad \text{Recall} = \frac{TP}{TP + FN + \epsilon} \tag{9}$$

  Where TP, FP, and FN represent true positive, false positive, and false negative counts across the spatial grid, and $\epsilon = 10^{-8}$ prevents division by zero.

- **Pixel-level F1:** Evaluates overall segmentation accuracy from the pixel-level precision and recall above:

$$\text{F1} = 2 \cdot \frac{\text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec} + \epsilon} \tag{10}$$

- **Boundary F1:** Restricts precision and recall to fire-perimeter only, obtained by morphological erosion of the predicted and ground truth masks. This isolates edge/shape accuracy from interior-region, which pixel-level F1 can mask:

$$\text{F1}_b = 2 \cdot \frac{\text{Prec}_b \cdot \text{Rec}_b}{\text{Prec}_b + \text{Rec}_b + \epsilon} \tag{11}$$

  where Prec_b and Rec_b are precision and recall computed only over boundary pixels of the predicted and ground truth masks.

- **AUPRC (Area Under the Precision-Recall Curve):** Measures the performance of the model across different thresholds:

$$\text{AUPRC} = \sum_k (R_k - R_{k-1}) P_k \tag{12}$$

  where P_k and R_k represent precision and recall evaluated at the k-th threshold operating point.

Due to the extreme rarity of the positive class, absolute metric values are difficult to interpret without a reference point. The following baselines were used:

1. an all-background predictor, assigning zero probability to every pixel.
2. a constant-prevalence predictor, assigning every pixel the measured test-set fire-pixel prevalence.

2. is the theoretically expected value of AUPRC under a classifier with no spatial discriminative ability, since a scorer that assigns identical scores to every pixel yields precision equal to prevalence at every recall level. Model performance is reported alongside both baselines.

5 Results

6 Discussions

A limitation of the cascaded pipeline is the use of Stage 2's intermediate predictions. Because true matching VIIRS/Landsat pairs are unavailable, Stage 2 is trained using Stage 1's predicted

VIIRS-resolution masks rather than true VIIRS observations. This intermediate target is model-derived rather than observational. A consequence is that any error in Stage 1 propagates into Stage 2's training, meaning Stage 2 also learns to downscale Stage 1's errors alongside fire information. This is a direct counterpoint to the motivation for cascaded downscaling: while cascading may reduce error from the resolution gap itself, it introduces source of error from stage-to-stage prediction dependency.

7 Conclusions

Future work should explore the integration of temporal information to capture the dynamic evolution of wildfires and/or predict burn areas. Another could include integration of climate data, and/or predicting fire intensity as well. Additionally, further research could investigate the use of generative models to enhance downscaled outputs and improve the model's ability to generalise across different geographic regions and fire regimes.

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